CutOut

What's it's all about?

Image Segmentation is a known problem in the world of Image Processing, where we try to partition a given image into several different segments that compose the original image. One use of the segmentation process is the separation of an object of interest from the rest of the image. Many algorithms use some form of user input in order to locate a specific object of interest and segment it out. An optional form of user inputs are scribbles, simple lines drawn by the user. The user is usually requested to draw at least one scribble inside the object of the interest and one on the outside, those two scribbles give the algorithm the basic information it needs in order to segment the object of interest from the rest of the image.

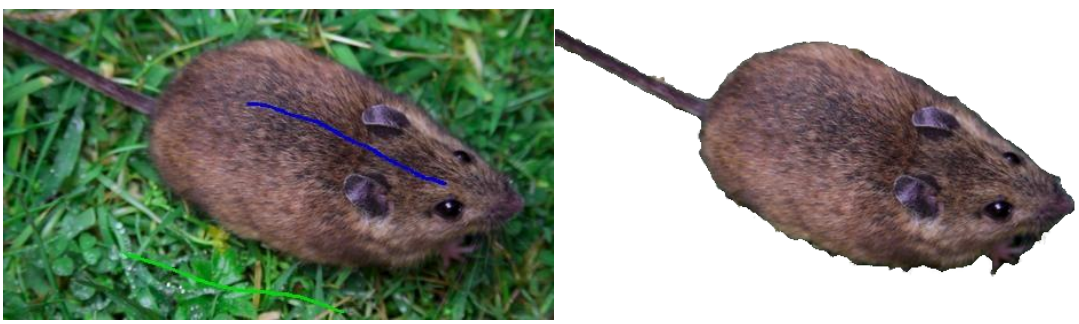
What did we do?

In our project we've implemented a scribble-based algorithm for extracting object from natural photos and pasting them seamlessly into a different background under the constrains of a mobile device's computational power. The algorithm was first developed and tested on a personal computer with the help of openCV's C++ libraries and was then ported to Android using the Native Development Kit. We've used the Android Software Development Kit in order to wrap the algorithm in a user friendly interface to create an application that anybody can use.

How does it work?

First, the user provides us with two (or more) scribbles, one is inside the object of interest while the other is in the background. The simple scribbles give us the starting point we need for separating the object from the background. Next, we propagate the information from the given scribbles to the rest of the image by calculating the weighted distance from each scribble to the rest of the pixels in the image. The proximity between pixels is defined using a smart weight function which is based upon each scribble's color profile. That way, the resulting distances are smaller for pixels that reside inside the object and higher for pixels that are part of the background. Thus, we can denote the pixels that are closer to the object's scribble as our object of interest and get the resulting segmentation. Another algorithmic step is then applied on the edges to refine our result and give each pixel a transparency value, resulting in a more natural looking segmentation.

Segmentation Result



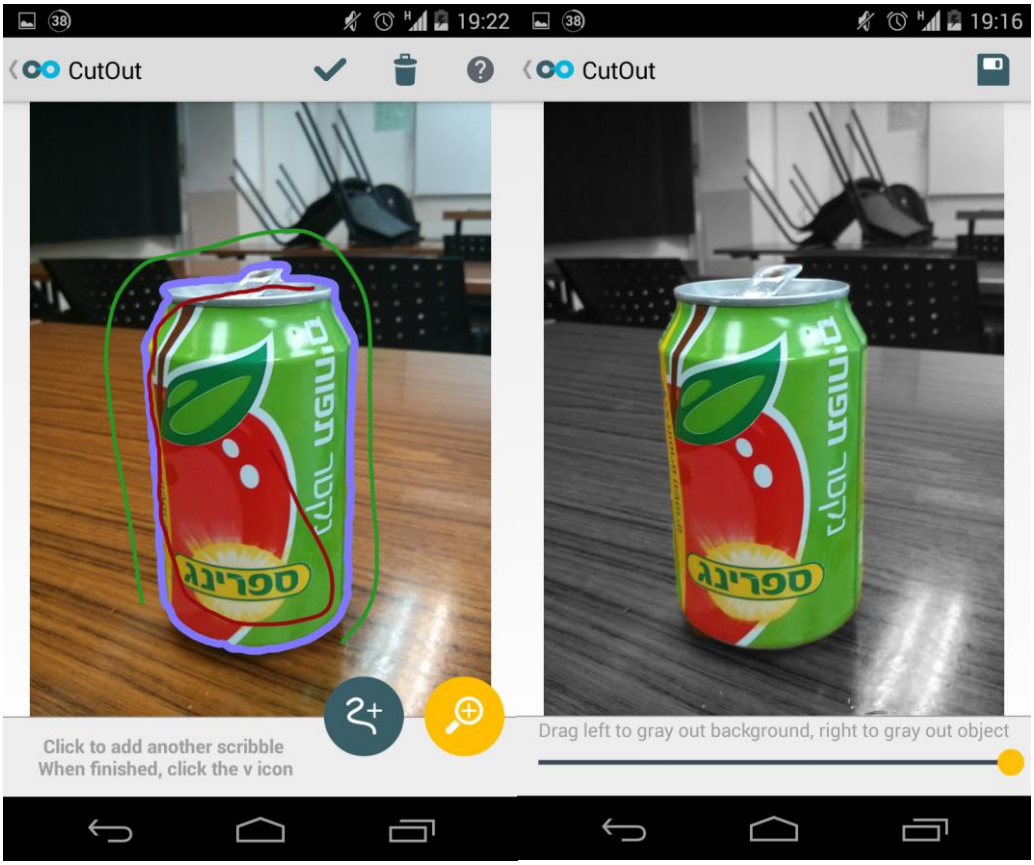
Where's the challenge?

Like many other algorithms in the world of Computer Vision, our basic segmentation algorithm is computationally heavy and cannot be ran in real-time on a mobile device. We've put a lot of effort into reducing the calculation time to mere seconds while still maintaining good segmentation results. The time reduction was done by running the major part of our algorithm on a scaled down image in order to get the rough segmentation, while later refining the segmentation's edges to get the final result. We've also experimented with different methods for profiling the scribbles and calculating the distances between pixels in order to find the algorithms that are optimal for our need in means of time and precision.

Who can use it?

Anyone! A major part of the project was to wrap our segmentation algorithm in a simple and intuitive application that anybody can use, thus bringing the power of our segmentation algorithm to the public.

Application Sample



About us

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